

Day-68

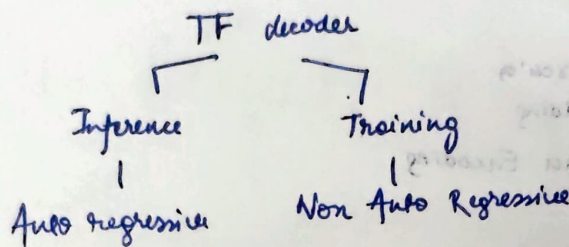
Lec-81

Masked Self Attention

Masked Multi-head Attention

Autoregressive models

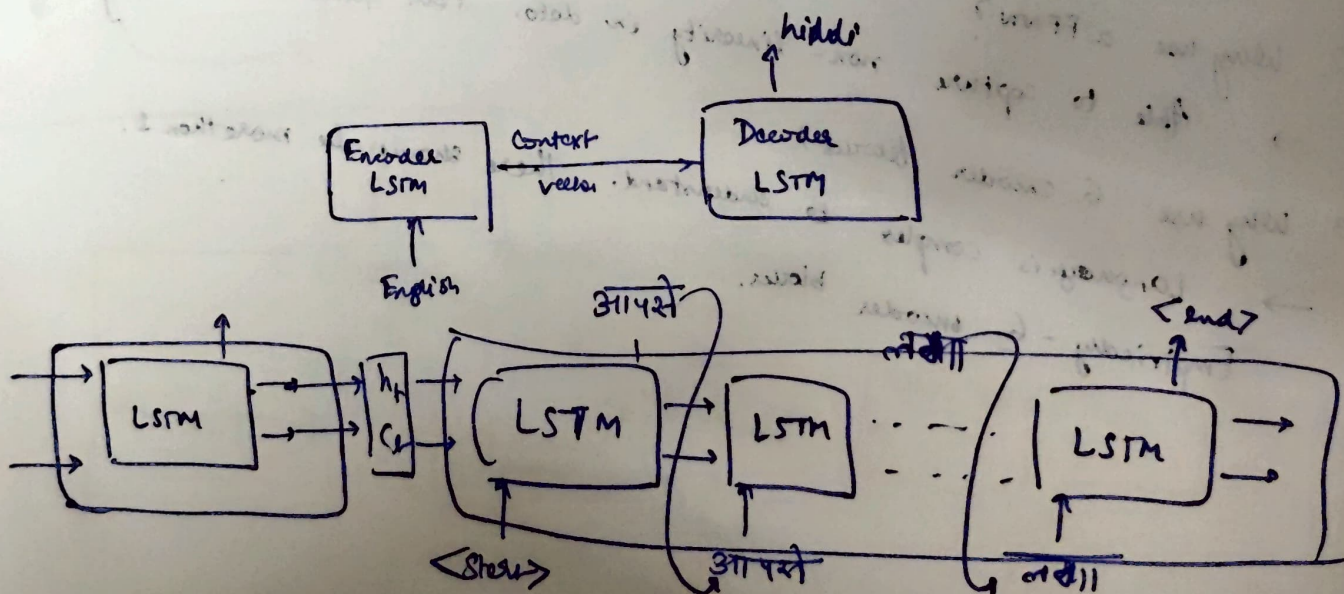
→ The transformer decoder is autoregressive at inference time & non-autoregressive at training time.



Autoregressive → Class of models that generate data points in a sequence by conditioning each new point on the previously generated points.

masked self attention

Basic Encoder - Decoder.



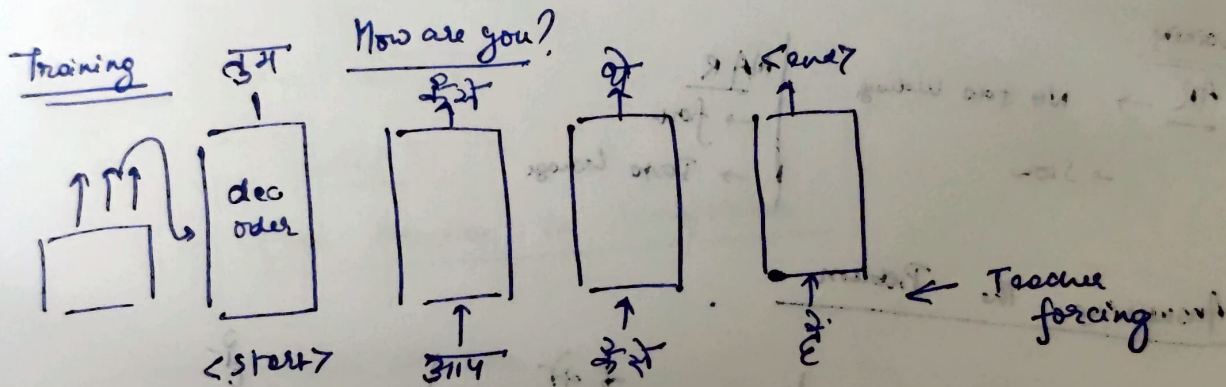
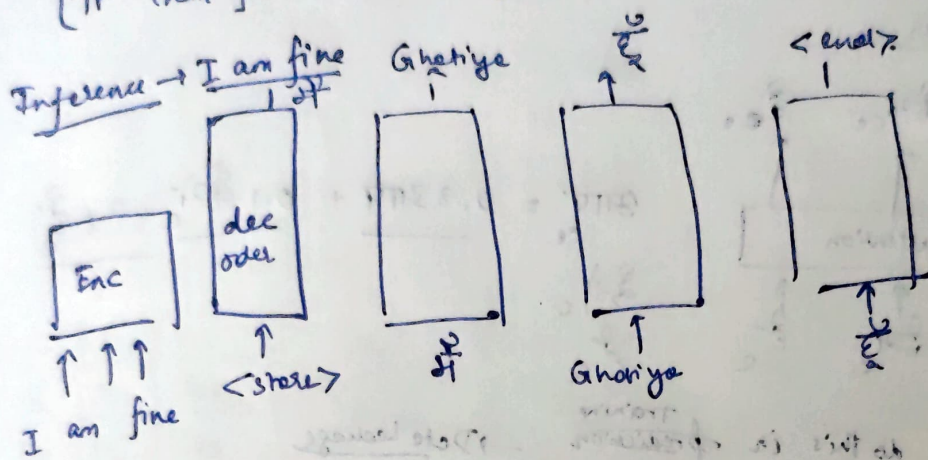
Auto regressive

Can't generate parallelly as it's sequential data.

Proof of concept

Let's assume $\rightarrow$ Inference Training $\rightarrow$ Autoregressive

[TF train] $\rightarrow$ inference



आप कैसे हैं $\rightarrow$ Loss function
 $\rightarrow$ Backpropagation

training $\rightarrow$ Autoregressive

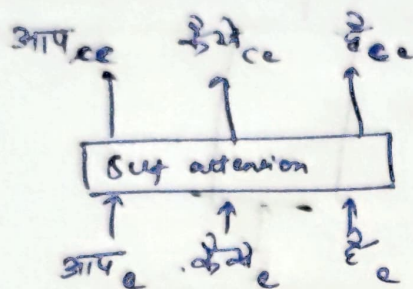
Problem? $\rightarrow$ 1. Slow as 4 operations for 1 row

Beaus of teacher forcing $\rightarrow$ no constraints for autoregressive

$\Rightarrow$ parallel execution
 $\Rightarrow$ faster execution.

The problem in parallelizing

Ques Training Encoder



$$\text{आप}_{ee} = \frac{0.8 \text{ आप} + 0.1 \text{ करे}_{ee} + 0.1 \text{ है}_{ee}}$$

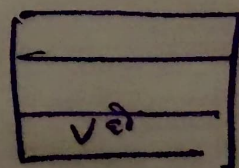
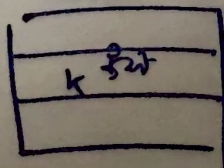
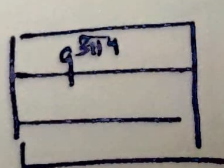
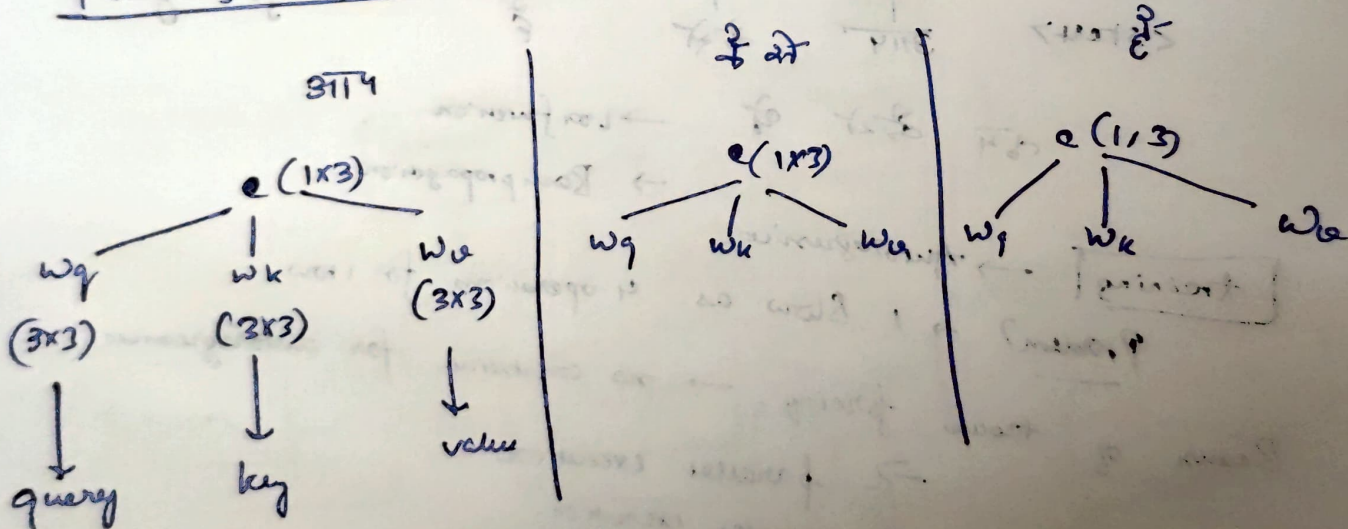
Can't do this in Training prediction → Data Leakage

Training

AR → No data leakage
→ Slow

MAR
→ fast
→ Data leakage

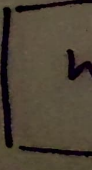
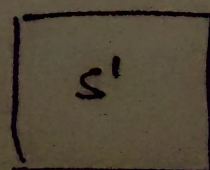
Answering the Problem



dot product

s_{11}	s_{12}	s_{13}
s_{21}	s_{22}	s_{23}
s_{31}	s_{32}	s_{33}

Scale
 $1/\text{sqrt}(d_k)$



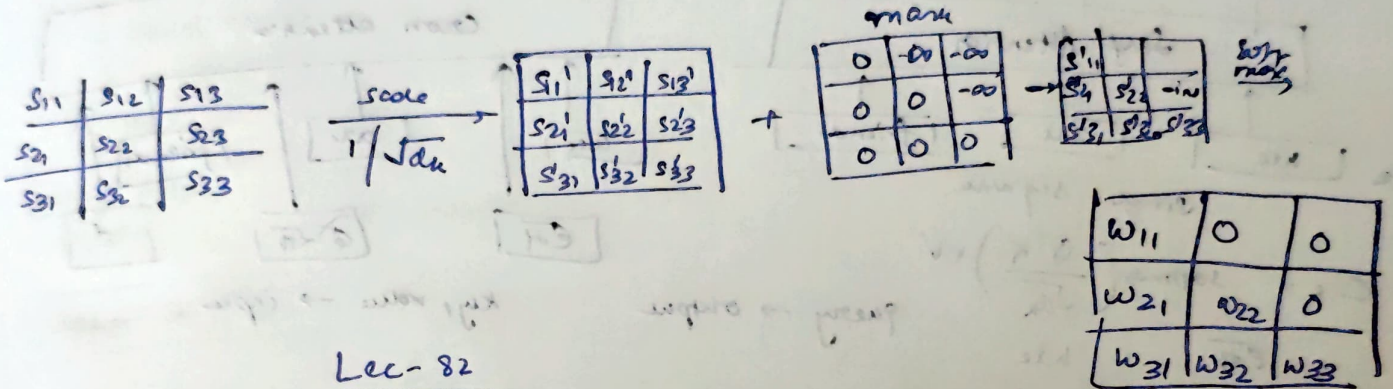
$$\mathbf{z}_{ce} = w_{11} \times v_{z_{ce}} + w_{12} \times v_{z_{ce}} + w_{13} \times v_{z_{ce}}$$

Contextual embedding

we don't need these contextual embeddings

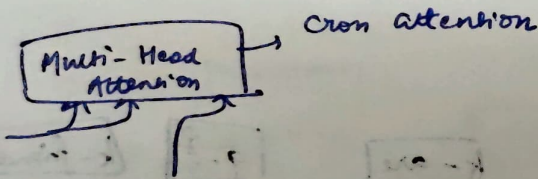
$$\mathbf{z}_{ce} = w_{21} + w_{22} + w_{23}$$

$$\mathbf{z}_{ce} = w_{31} + w_{32} + w_{33}$$



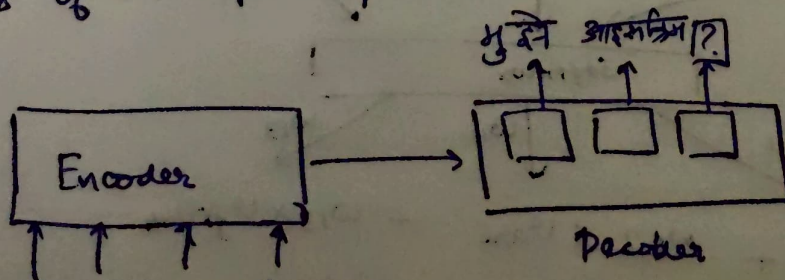
Lec-82

Cross Attention in Transformers



What is cross attention?

Cross attention is a mechanism used in transformer architectures, particularly in tasks involving sequence-to-sequence data like translation or summarization. It allows a model to focus on different parts of an input sequence when generating an output sequence.



I like eating icecream
(pardes)

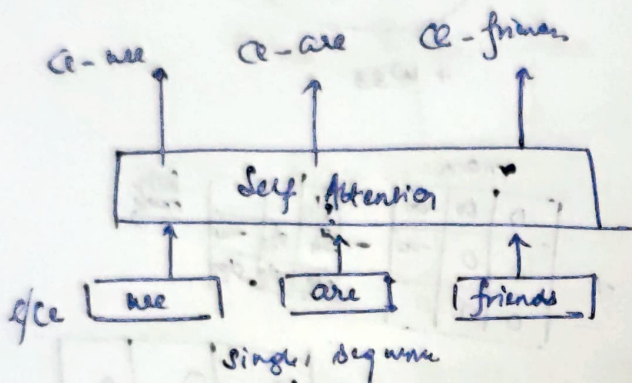
मुझे आइसक्रीम खाना पसंद है

like	0	.	.	.
eat	.	.	.	0
icecream	.	.	0	.

- 1) $\mathbf{z}_{ce}$ depends on:-
- 1) What has been generated till now (self attention)
 - 2) What is there in input sequence.
- Cross attention

Conceptually similar to Self-Attention

1. The input
2. the processing
3. The output

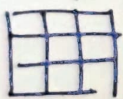


$$C_e = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) \times V$$

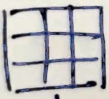
E_H

$e-E_H$

W_q



W_k



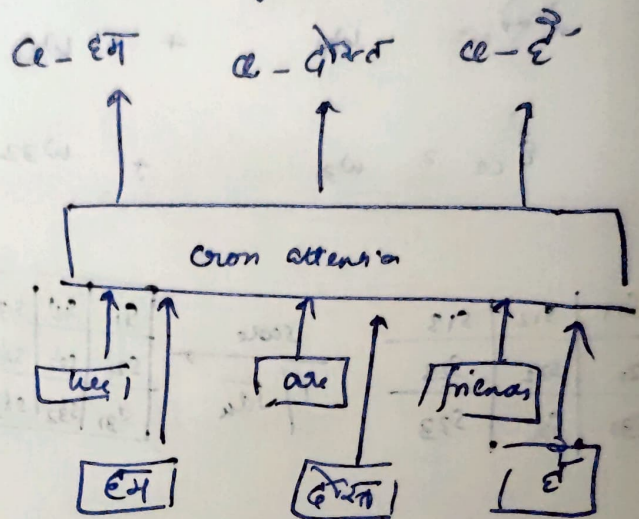
$q-are$

$k-are$

$v-are$

query → output

key, value → input



same no. of vectors as output sequence.

q

$q-E_H$

$k-are$

$v-are$

$q-E^p$

$k-are$

$v-are$

$q-E^p$

$k-friends$

$v-friends$

$q-E_H$ S_{11}

$q-E_H$ S_{12}

$q-E_H$ S_{13}

$k-are$

$k-are$

$k-friends$

softmax

w_{11}

w_{12}

w_{13}

$w_{11} \times q_{are}$

$+ w_{12} \times q_{are}$

$+ w_{13} \times q_{friends}$

↓

q-GNN

k-~~all~~, k-qae, k-friends

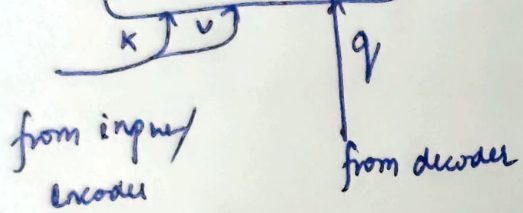
↓ softmax

$$w_{21} \times V_{koe} + w_{22} \times w_{koe} + w_{23} \times w_{friends}$$

we are friends

en	0	0	0
qae	0	0	0
friends	0	0	0

Multi-head Attention



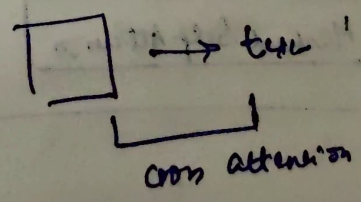
Output Difference

• Similar to

~~Bahdanau~~ Bahdanau Attention | Luong Attention

Use-cases

1. Machine Translation.
2. Image Captioning.



text → image

text → speech